An Exploratory Analysis of the Correlation Between GC Content and Optimal Growth Temperature in Restricted Temperature Ranges

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1. Abstract

GC pairs in a genome are known for their durability and stability. Based on this concept, higher GC content tends to be linked to DNA's durability in high-temperature environments. However, the exact correlation is still in ongoing debate; there are other factors that also influence the adaptability of organisms in high-temperature environments. This study examines the correlation between GC content and optimal growth temperature across 40 bacteria (20 mesophilic and 20 thermophilic), GC content differences in both groups, and reanalyzes a previous study that shows low to moderate positive correlation in its original range and restricted temperature range. While many previous studies with larger data sets have reported a positive correlation, in this study, a weak negative correlation is observed, possibly reflecting a restricted temperature range, sample size, phylogenetic bias, and other factors that contribute to thermostability. Using an independent sample t-test, this study shows no significant difference ($p > 0.05$) between GC content in both groups, while Pearson correlation methods showed there is a statistically significant negative correlation between GC content and optimal growth temperature in this data set ($r \approx -0.32$, $p \approx 0.045$). This study implies that observed positive correlation may depend on sampled temperature range, and the positive correlation between optimal growth temperature and GC content may not be consistently observed in certain data sets or smaller data sets.

2. Introduction

In DNA, Guanine (G) is paired with Cytosine (C), and Thymine (T) is paired with Adenine (A). A GC pair formed 3 hydrogen bonds, while an AT pair formed 2 hydrogen bonds. This trait has been studied thoroughly, and many studies suggests that GC pairs are associated with higher durability and thermostability mechanisms. Despite GC content correlation with optimal growth temperature, GC content is highly preserved in evolutionary lineage; some phylogenies tend to have higher GC content despite their growing environments (such as Streptomyces, Mycobacterium, and Actinomyces). Genomic GC content varies across organisms, ranging from 13% to 75% in some archaea. The variation of GC content is also influenced by mutation bias, natural selection, and biased gene conversion.

The debate regarding the correlation between GC content and optimal growth temperature often revolves around whether higher GC content in organisms (most studies using prokaryotes as samples) directly influences organisms' durability in extreme or moderately high temperature environments. Many studies have proved that GC content in certain regions, such as genes that produce rRNA and tRNA, has a

positive correlation with optimal growth temperature. However, the relation between the whole genomic GC content and optimal growth temperature has yielded inconsistent results.

While some studies that observe the relation between whole genomic GC content and optimal growth temperature indicate there is a low to moderate positive correlation, the results really depend on the sample size and temperature range. In other studies, the positive correlation disappears, possibly due to the limited data set. However, these conflicting results may also arise because thermal adaptation depends on multiple factors beyond GC content, including phylogenetic history, sample size limitations, membrane adaptations, and chaperone-mediated protein stability.

Another possible explanation for this inconsistency is that the correlation in a broad dataset might be driven by extreme organisms such as hyperthermophiles and psychrophiles, whereas in a narrowed dataset the correlation might weaken or completely disappear, as most previous studies that observe this correlation only use a wide range of optimal growth temperatures (10-120°C). Therefore, this study aims to test whether the reported association between genomic GC content and optimal growth temperature remains detectable when analysis is restricted to mesophilic and thermophilic bacteria (20–85°C). We hypothesize that the correlation between genomic GC content and optimal growth temperature weakens when analysis is restricted to mesophilic and thermophilic bacteria.

3. Methods

3.1 Dataset Collection

Forty bacteria representing mesophilic and thermophilic with optimal growth temperature ranges from 20 to 85 are selected. Mesophiles were defined as organisms with optimal growth temperature below 45°C and thermophiles were defined as organisms with optimal growth temperature above 45°C For each bacteria, genomic length, GC content, gram type, nucleotide frequency, and optimal growth temperature were recorded. All genome sequences are extracted from the National Center for Biotechnology Information, and other data regarding optimal growth temperature are obtained from BacDive. In this dataset, 20 mesophilic bacteria and 20 thermophilic bacteria were included.

3.2 GC Content and Nucleotide Frequency Calculation

GC content is the proportion of guanine and cytosine bases out of the four total bases. GC content percentage is calculated as:

$$\frac{G+C}{A+T+G+C} \times 100\%$$

While nucleotide frequency calculates the proportion of each bases out of the four total bases. Nucleotide frequency is calculated as:

$$\%BP = \frac{\text{number of sequences containing the noted base pair}}{\text{total number of sequences}} \times 100\%$$

This study used Python to calculate the whole genome length, nucleotide frequency, and GC content.

3.2 Statistical Analysis

A. Comparison between mesophiles and thermophiles' GC content

Welch's two-sample t-test:

H_0: Mean GC mesophiles = Mean GC thermophiles

H_1: There is a statistically significant difference in mean GC content between mesophiles and thermophiles

Significance: $\alpha = 0.05$

B. Correlation analysis

The correlation between GC content and optimal growth temperature is measured with Pearson correlation:

$$r = \frac{\sum(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum(X_i - \overline{X})^2 \sum(Y_i - \overline{Y})^2}}$$

3.3 Reanalysis of previous study

The dataset from previous study was reanalyzed under two conditions:

1. Full temperature range (0-120°C)
2. Restricted temperature range (20-85°C)

Pearson correlation applied to both conditions.

Data points were extracted from a published figure in Hu et al. (2022) using WebPlotDigitizer and reanalyzed in Python.

3.4 Computational Implementation

All analyses were performed in Python using pandas for data handling, scipy.stats for statistical tests, and plotly for visualization.

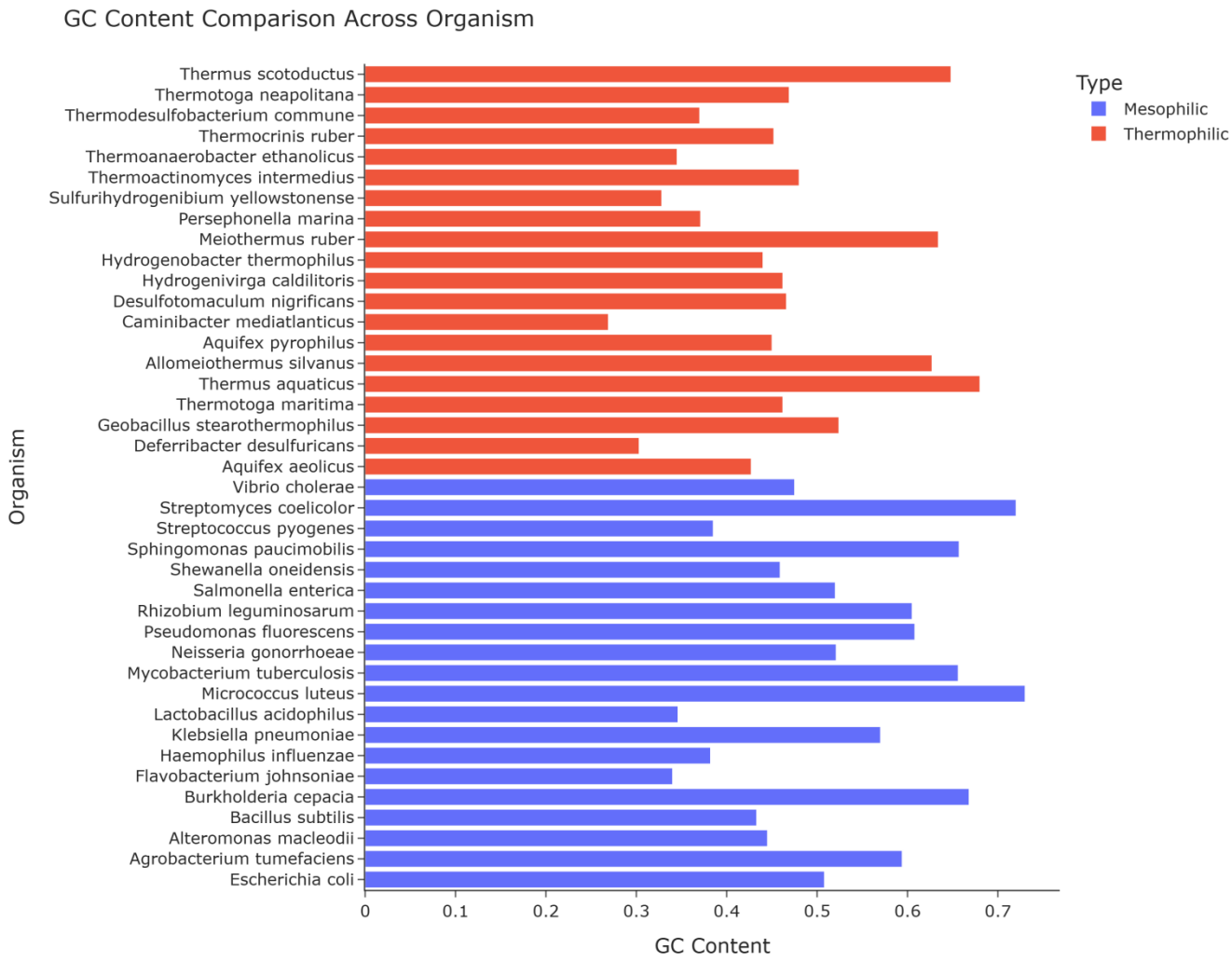
4. Results

4.1 Table Results

Organism	Genome length	GC Content	Frequency of Adenine	Frequency of Thymine	Frequency of Guanine	Frequency of Cytosine	Type	Optimal growth temperature (in Celsius)	Gram type
Escherichia coli	4641652	0.508000	0.246000	0.246000	0.254000	0.254000	Mesophilic	37	Negative
Agrobacterium tumefaciens	5511883	0.594000	0.204000	0.202000	0.296000	0.299000	Mesophilic	28	Negative
Alteromonas macleodii	4867393	0.445000	0.277000	0.277000	0.223000	0.222000	Mesophilic	22	Negative
Bacillus subtilis	4226648	0.433000	0.283000	0.284000	0.217000	0.217000	Mesophilic	37	Positive
Burkholderia cepacia	8366868	0.668000	0.166000	0.166000	0.334000	0.333000	Mesophilic	35	Negative
Flavobacterium johnsoniae	6041200	0.340000	0.329000	0.331000	0.169000	0.172000	Mesophilic	25	Negative
Haemophilus influenzae	1890662	0.382000	0.308000	0.310000	0.190000	0.192000	Mesophilic	37	Negative
Klebsiella pneumoniae	5711891	0.570000	0.215000	0.215000	0.287000	0.283000	Mesophilic	37	Negative
Lactobacillus acidophilus	1965390	0.346000	0.327000	0.327000	0.172000	0.174000	Mesophilic	37	Negative
Micrococcus luteus	2478263	0.730000	0.135000	0.135000	0.365000	0.364000	Mesophilic	30	Positive
Mycobacterium tuberculosis	4411532	0.656000	0.172000	0.172000	0.327000	0.329000	Mesophilic	37	Positive
Neisseria gonorrhoeae	2292467	0.521000	0.239000	0.240000	0.261000	0.260000	Mesophilic	37	Negative
Pseudomonas fluorescens	6610571	0.608000	0.196000	0.196000	0.304000	0.304000	Mesophilic	28	Negative
Rhizobium leguminosarum	8567819	0.605000	0.198000	0.196000	0.300000	0.305000	Mesophilic	28	Negative
Salmonella enterica	4723925	0.520000	0.240000	0.240000	0.260000	0.260000	Mesophilic	37	Negative
Shewanella oneidensis	5131416	0.459000	0.271000	0.270000	0.230000	0.229000	Mesophilic	33	Negative
Sphingomonas paucimobilis	4367078	0.657000	0.171000	0.172000	0.329000	0.328000	Mesophilic	30	Negative
Streptococcus pyogenes	1826467	0.385000	0.310000	0.306000	0.195000	0.189000	Mesophilic	37	Positive
Streptomyces coelicolor	9054847	0.720000	0.140000	0.140000	0.360000	0.360000	Mesophilic	30	Positive
Vibrio cholerae	4007146	0.475000	0.263000	0.261000	0.237000	0.238000	Mesophilic	35	Negative
Aquifex aeolicus	1125385	0.427000	0.286000	0.286000	0.213000	0.214000	Thermophilic	85	Negative
Deferribacter desulfuricans	2542933	0.303000	0.348000	0.349000	0.152000	0.151000	Thermophilic	62	Negative
Geobacillus stearothermophilus	2790037	0.524000	0.238000	0.238000	0.265000	0.258000	Thermophilic	62	Positive
Thermotoga maritima	1869644	0.462000	0.268000	0.270000	0.228000	0.235000	Thermophilic	80	Negative
Thermus aquaticus	2338641	0.680000	0.161000	0.159000	0.339000	0.342000	Thermophilic	70	Negative
Allomeiothermus silvanus	3721669	0.627000	0.186000	0.187000	0.314000	0.313000	Thermophilic	55	Negative
Aquifex pyrophilus	1624701	0.450000	0.273000	0.277000	0.225000	0.225000	Thermophilic	85	Negative
Caminibacter mediatlanticus	1685887	0.269000	0.368000	0.364000	0.137000	0.132000	Thermophilic	55	Negative
Desulfotomaculum nigrificans	2892255	0.466000	0.267000	0.267000	0.233000	0.233000	Thermophilic	58	Positive
Hydrogenivirga caldilitoris	1753429	0.462000	0.271000	0.267000	0.232000	0.230000	Thermophilic	75	Negative
Hydrogenobacter thermophilus	1742932	0.440000	0.278000	0.282000	0.219000	0.221000	Thermophilic	73	Negative
Meiothermus ruber	3097457	0.634000	0.183000	0.183000	0.318000	0.316000	Thermophilic	60	Negative
Persephonella marina	1983964	0.371000	0.310000	0.319000	0.189000	0.182000	Thermophilic	73	Negative
Sulfurihydrogenibium yellowstonense	1534471	0.328000	0.338000	0.334000	0.166000	0.162000	Thermophilic	70	Negative
Thermoactinomyces intermedius	2756563	0.480000	0.259000	0.261000	0.236000	0.245000	Thermophilic	50	Positive
Thermoanaerobacter ethanolicus	2827640	0.345000	0.322000	0.332000	0.171000	0.174000	Thermophilic	70	Positive
Thermocrinis ruber	1521037	0.452000	0.279000	0.269000	0.229000	0.223000	Thermophilic	80	Negative
Thermodesulfobacterium commune	1764045	0.370000	0.316000	0.314000	0.186000	0.184000	Thermophilic	70	Negative
Thermotoga neapolitana	1884562	0.469000	0.265000	0.266000	0.233000	0.236000	Thermophilic	77	Negative
Thermus scotoductus	2069303	0.648000	0.176000	0.176000	0.326000	0.322000	Thermophilic	65	Negative

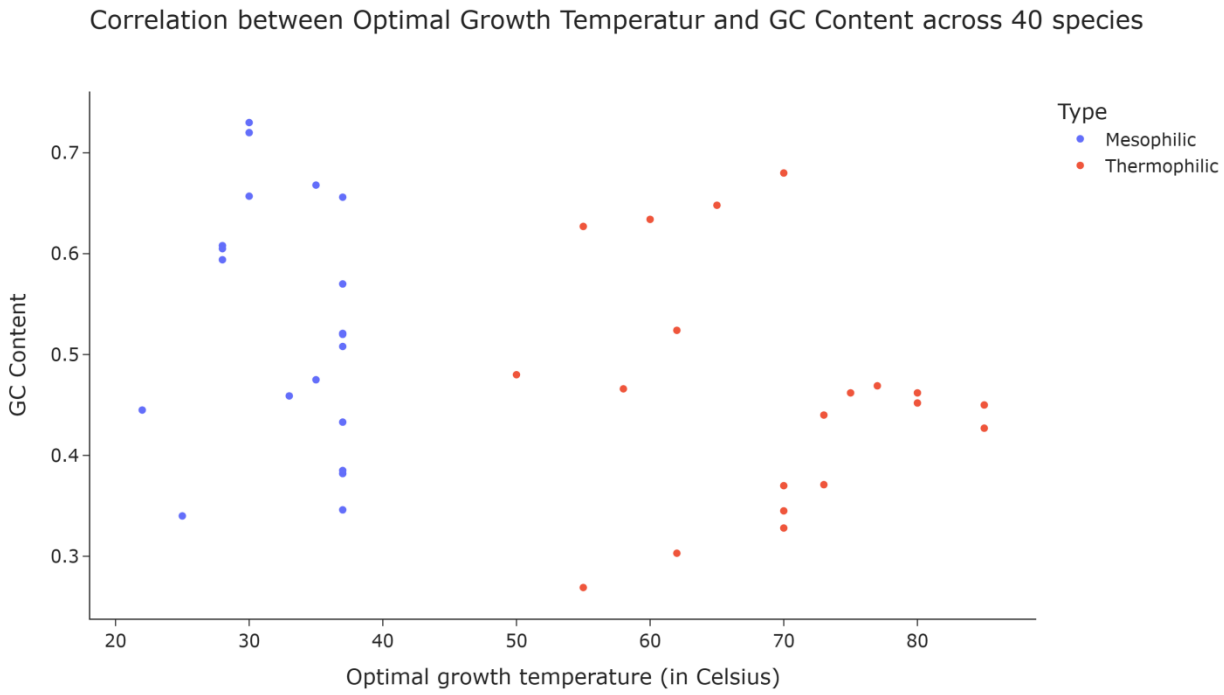
This table shows all the data recorded for each bacterium including genome length, nucleotide frequency, GC content proportion, organism type, its optimal growth temperature, and gram type.

4.2 GC Content Across 40 Bacteria



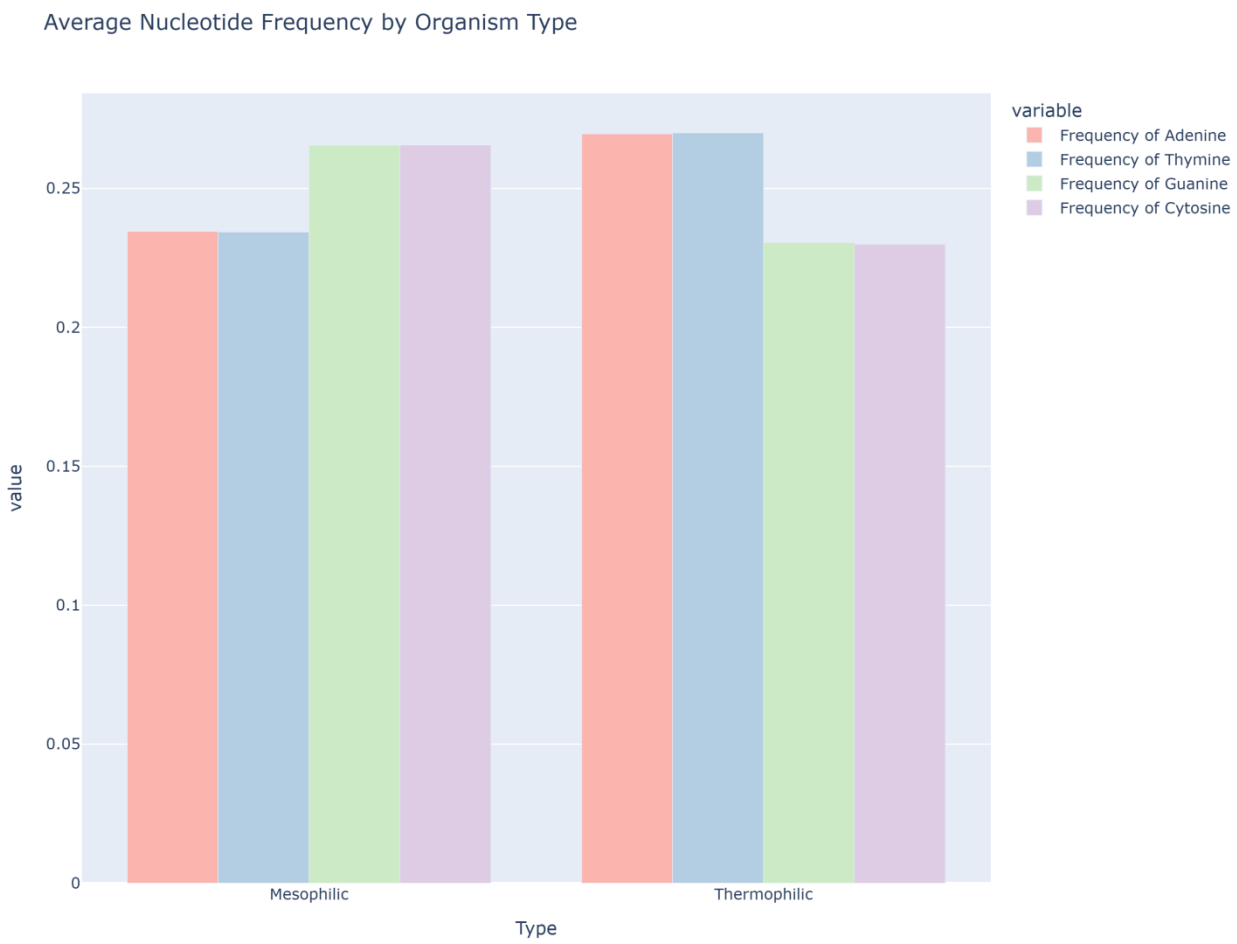
This figure shows the GC content comparison across 40 bacteria, with red color representing thermophilic bacteria and blue color representing mesophilic bacteria.

4.3 Correlation Between GC Content and Optimal Growth Temperature



This scatter plot represents the correlation between optimal growth temperature and GC content across 40 species, with blue dots indicating mesophilic bacteria and red dots indicating thermophilic bacteria.

4.4 Average Nucleotide Frequency in Mesophiles and Thermophiles



This group bar chart shows the difference in average nucleotide frequency in mesophiles and thermophiles. With red representing average adenine frequency, blue representing average thymine frequency, green representing average guanine frequency, and purple representing average cytosine frequency.

4.5 Statistical Results

Welch's two-sample t-test between mesophiles and thermophiles:

T-test statistic: 1.879

T-test p-value: 0.068 (insignificant)

Pearson correelation:

Pearson correlation (r): -0.318

Pearson p-value: 0.045 (significant)

4.6 Reanalysis of Previous Study

Reanalysis of the data reported by Hu et al. (2022) across the full temperature range showed a positive correlation between GC content and optimal growth temperature ($r = 0.2832$, $p = 0.0012$, $n = 128$). However, under the restricted temperature range (20–85°C), the correlation weakened ($r = 0.0217$, $p = 0.8436$, $n=85$).

5. Discussion

5.1 Relation Between GC Content and Thermophily

The finding of this study suggests that genomic GC content is not sufficient to explain organism survival in a thermal environment. Although higher GC content is often associated with higher DNA stability, including elevated DNA melting point. The observed significant negative correlation suggests that higher genomic GC content may not universally contribute to thermophilic adaptation, and may reflect the influence of other biological or sampling factors. Previous studies have suggested that enzyme and protein stability, chaperone-assisted protein folding, and the enhanced stability of structural RNAs such as tRNA and rRNA may play more direct roles in adaptation to high-temperature environments.

While GC content is proven to increase the melting temperature of DNA, the correlation of whole genomic GC content and organisms' survival in high-temperature environments remains controversial. This question remains controversial as there are more significant factors that contribute to organisms' adaptation in high-temperature environments besides genomic GC content, such as enzyme and protein stability, RNA stability, and chaperone protein adaptation. Enzyme and protein stability play an important role in organisms' survival in thermal environments, as they assist almost every reaction in the cell. Enzymes and proteins tend to be sensitive towards pH and temperature, thus making them one of the most vulnerable components in a cell. In extreme environments, a protein may misfold, making it ineffective. This scenario is where chaperone adaptability plays an important role in the thermostability mechanism. Molecular chaperones may contribute to thermotolerance by promoting correct protein folding, refolding denatured proteins, and preventing heat-induced aggregation. All this mechanism is crucial for organisms' stability in a heat-stress environment. And lastly, RNA stability tends to be more crucial than whole genomic stability, as RNA is less stable than DNA and particularly susceptible to thermal denaturation. Studies have also shown that higher GC enrichment in tRNA and rRNA correlates with organismal growth temperature, suggesting RNA stability may contribute to thermophilic adaptation. Together, these mechanisms may help explain why genomic GC content alone may be an incomplete predictor of thermophily.

5.2 Reanalysis of Previous Study

Although Hu et al. (2022) reported a significant positive correlation between whole genomic GC content and optimal growth temperature in a broad temperature range and large dataset (0-120°C, n = 150), reanalysis of their digitized data using Pearson correlation analysis (with temperature range limited to 20-85°C, n = 85) revealed a positive but statistically non-significant correlation ($p = 0.8346$). Consistent with the complexity of this relationship, the present study showed a weak but statistically significant negative correlation between GC content and optimal growth temperature. (n = 40). In addition, through Welch's t-test, the present study shows there is no statistically significant difference between mesophiles' and thermophiles' GC content. Collectively, all these findings suggest that a positive correlation between whole genomic GC content and optimal growth temperature may be dependent on sample range, phylogenetic bias, and sample size.

5.3 Limitation

This study is still limited by sample size (n = 40) that might limit statistical power and increase phylogenetic bias during the sampling. Phylogenetic bias in this study may also influence the overall statistic, as the high GC content might not be caused by thermal adaptation but by the evolutionary effect. In this sample, there are several organisms descending from a genus known for its high GC temperature, such as Streptomyces. In addition, a whole genome may be too broad to analyze compared to specific regions such as coding regions or RNA.

5.4 Future Direction

Future study using a larger data sample and phylogenetically controlled analysis may provide a more definitive answer about whether GC content in the whole genome correlates with organisms' survival in thermal environments. In addition, observing certain regions such as coding regions and promoter GC content, codon usage bias, and RNA structure differences in mesophilic and thermophilic organisms may provide more insight about the thermostability mechanism rather than the whole genome analysis.

6. Reference

The previous study was cited from En-Ze Hu, Xin-Ran Lan, Jie Gao, and Deng-Ke Niu. A positive correlation between GC content and growth temperature in prokaryotes. 2022 Feb 9;23:110. doi: 10.1186/s12864-022-08353-7. The complete genomic sequences for all analyzed species were obtained from the National Center for Biotechnology Information, while the optimal growth temperatures were obtained from BacDive. In addition, the significance of chaperones in thermostability is cited from Xu Li et al. (2025).